Competitive Dynamics in Organizations

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Abstract

Existing theories have identified many preconditions for sustained economic growth, yet few explore the endogeneity of the incentive problem, which is of central importance to understanding productivity. We develop a theory of competitive dynamics that explains how incentive systems are endogenously formed and why productivity diverges across institutional environments. We assert that nations and regions adopt distinct incentive levels because the modes of maintaining internal organizational authority differ fundamentally. Because contracts are inherently incomplete, authority cannot be sustained solely through formal legal institutions and must also be maintained through politically strategic actions. When legal enforcement is strong, authority is predominantly contractual and merit-based competition is strengthened. When legal enforcement is weak, authority relies more heavily on individualized loyalty, creating a trade-off between preserving political control and sustaining high-powered incentives—though extensive loyalty networks can still support strong competition. This mechanism not only reinforces the conventional view that strong legal institutions promote economic growth but also provides a microfoundation for both sustained high and low productivity under weak legal enforcement. By endogenizing authority maintenance, our theory provides a unified microfoundation for endogenous incentive formation, persistent productivity divergence, politically structured inequality, and institutional change.

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Introduction

Why do some countries consistently sustain high-powered, performance-based competition while others fail to do so? Variation in the weight placed on meritocracy not only cuts across regime type and development level but also shifts markedly within countries over time. The United States has maintained a high-powered incentive system for more than a century. West Germany's postwar economic miracle, Singapore's development model, and South Korea's industrial takeoff likewise reflect prolonged reliance on strong performance-based incentives. In contrast, Haiti and the Democratic Republic of the Congo have experienced extended periods of low-powered incentives and stagnation, while Argentina has followed a highly volatile trajectory of repeated booms and busts. Explaining why countries sustain different incentive intensities is therefore central to understanding productivity divergence.

This paper develops an organizational-economics framework for analyzing how competitive dynamics are formed and sustained within organizations. Large-scale productive activity is rarely sustained by isolated individuals; it is typically carried out within organizations and requires coordination. Coordination depends on the establishment of authority, which consists in subordinates' willingness to comply with superiors' directives. Where the rule of law is strong, coordination is primarily governed by contracts: compliance is backed by impersonal legal enforcement, making authority largely contractual. When legal enforcement is weak, contractual authority loses effectiveness and compliance increasingly depends on surbordinates' loyalty. Superiors therefore rely more heavily on political strategies, with favoritism serving as a mechanism for sustaining control by allocating promotions and resources to loyal subordinates. Because contracts are inherently incomplete, organizational authority always carries a political dimension, though institutions determine its extent. Our framework highlights two mechanisms that jointly shape the intensity of internal competition: (i) the legal foundation of authority, which conditions reliance on favoritism to sustain authority politically, and (ii) the structure of loyalty networks, which determines how easily authority can be maintained through political means. Their interaction generates systematic productivity differences across organizations and institutional environments.

In relatively strong legal environments, organizational authority is more likely to rest on legally enforceable contracts. By specifying superiors' directives and subordinates' responsibilities in formal terms, contracts provide an impersonal basis of authority backed by state enforcement. This impersonal authority reduces reliance on favoritism by lessening superiors' dependence on the personal loyalty of particular subordinates for internal coordination. As a result, when selecting candidates for positions, superiors are more likely to adopt stricter, more meritocratic competitive mechanisms, thereby sustaining stronger incentives and higher organizational output.

Counterintuitively, weaker legal environments do not necessarily imply lower economic performance. When legal enforcement is weak, contracts provide limited support for coordination, and authority must rely more heavily on non-coerced compliance rooted in loyalty. Favoritism becomes more prevalent, but its implications depend on the structure of the loyalty network. Large loyalty networks enable superiors to sustain high-powered competition even under weak legal enforcement. When a high share of competitors are loyal, meritocratic winners are still likely to be loyal, so strict competition does not threaten authority. In this case, superiors can adopt intensive performance-based competition without risking the empowerment of disloyal agents, thus generating stronger incentives and higher output.

Weak competition arises only when both legal enforcement and loyalty are low. With weak contractual authority accompanied by few loyal subordinates, strict meritocracy risks allocating key positions and resources to disloyal agents, a risk that fragile contractual authority cannot absorb. To preserve control, superiors shift weight toward loyalty in promotions and resource allocation, weakening performance-based incentives and reducing organizational productivity.

These two mechanisms are consistent with broad historical and contemporary patterns. Strong rule-of-law institutions are one foundation for high-powered incentives, but they are not the only one. In many historical cases such as the Ottoman Empire's centralized fiscal-administrative system (Karaman and Pamuk 2010; Pamuk 2020), Meiji Japan's state-led modernization (Johnson 1982), Prussia's administrative and market-oriented reforms (Dincecco 2011; Andersen 2018), and imperial China's bureaucratic coordination (Chen 2024; Wang 2022), large-scale economic mobilization emerged before modern legal institutions were fully developed, supported instead by centralized political authority and bureaucratic capacity. Modern cases show the same logic. China's reform-era promotion system tied local officials' careers to economic performance while maintaining political loyalty as the basis of authority (Jia et al. 2015). At the firm level, evidence on family CEO succession (Pérez-González 2006; Bennedsen et al. 2006) and cross-country differences in management practices (Bloom and Van Reenen, 2007, 2010) shows that internal allocation rules based on family ties strongly shape productivity.

The heterogeneous incentives that subordinates face, shaped by favoritism and by the structure of informal loyalty networks, are a key factor not only in generating aggregate productivity differences across institutional contexts but also in shaping broader economic inequality in our framework. When competition is weakening, equilibrium effort decreases for all subordinates, but loyal subordinates choose higher effort than disloyal subordinates. This is because loyal subordinates have a weakly higher probability of being promoted to key positions or receiving valuable organizational resources than disloyal subordinates. This can produce persistent heterogeneity in effort and reward over time. Such politically induced differences in incentives constitute a central mechanism through which authority structures generate persistent economic inequality.

We also discuss how democratic backsliding may arise from the accumulation of political authority, which is reflected in superiors' consistent willingness to expand loyal networks and to selectively weaken institutional constraints, and how this process can trigger significant disruptions in economic performance. What are the implications of leadership changes, through succession, removal, or mortality, for economic performance and for institutional design aimed at sustaining long-term stable economic growth. Finally, our framework helps reconcile one of the central puzzles in comparative political economy: why countries with similar regime types often exhibit dramatically different long-run growth trajectories, while countries under different regime types sometimes achieve remarkably similar economic performance. [1]

Although many factors contribute to productivity divergence, the maintenance of authority is fundamental. Without an appropriate incentive structure, preconditions emphasized in economics—such as resource endowments (Sachs and Warner 1995, advanced technology (Solow 1957; Romer 1990), and formal property rights (North 1990; Acemoglu et al. 2001)—do not automatically translate into productive activity. Indeed, technological upgrading and strengthened enforcement are often endogenous outcomes, visible manifestations of periods of intense underlying incentives. A system endowed with sophisticated technology can still stagnate if resource allocation ceases

[1]A longstanding debate in political economy concerns whether democracy promotes economic growth. Existing evidence remains inconclusive. Some studies find that democratic institutions improve long-run economic performance by strengthening accountability and property rights (Lipset 1960; Persson and Tabellini 2006; Acemoglu et al. 2019), whereas others report negative (Huntington 1973; Tavares and Wacziarg 2001; Mobarak 2005) or statistically insignificant effects (Przeworski 2000; Gerring et al. 2005). More importantly, even studies documenting positive average effects struggle to reconcile their conclusions with prominent counterexamples, such as the sustained rapid growth of authoritarian economies including China.

Our framework suggests that this ambiguity arises because regime type itself is not the fundamental determinant of productive performance. Democratic regimes differ substantially in the extent to which the rule of law enables contractual authority, while authoritarian regimes vary in their capacity to sustain political authority through stable organizational control. Consequently, both democracies and autocracies can achieve sustained high productivity when they possess a sufficiently strong form of authority—contractual or political—to support high-powered incentives. Conversely, when neither source of authority is sufficiently developed, leaders must weaken internal competition to preserve organizational stability, reducing productive efficiency regardless of regime type. By shifting attention from regime labels to the institutional foundations of authority, our framework reconciles the mixed empirical findings in the existing literature.

to be meritocratic. The reason is straightforward: economic incentives cannot operate in a vacuum. All productive activity presupposes political order.[2] Before individuals can invest, cooperate, or compete, they must be coordinated through institutions that secure compliance and stabilize expectations. Production is therefore inherently political. Markets, firms, and bureaucracies appear purely economic, but none can function without an underlying authority structure that anchors behavior. The central question of development is thus not merely what conditions make growth possible, but what authority structures enable a system to sustain the incentives that generate growth.

Our work connects to three primary strands of literature across economics and political science. First, our paper speaks to the landmark institutional economics literature, which posits that property rights, the rule of law, and executive constraints are the fundamental drivers of long-run economic divergence (North 1992; Greif 2006; Acemoglu et al. 2001; Acemoglu and Robinson 2005). Although this literature emphasizes the institutional impact on economic performance, it neither discusses growth mechanisms under weak legal enforcement nor offers a unified logic for growth episodes across varied institutional contexts, which is central to our framework.

Second, our paper extends the understanding of authority and contributes to incentive theory within organizational economics. While traditional incentive theories treat authority as a ubiquitous endowment where contractual compliance is guaranteed and principals can effortlessly deploy tournaments, their analytical scope is largely restricted to optimal incentive design within well-functioning legal regimes (Coase 1937, Holmström 1982; Baker et al. 1994, Lazear and Rosen 1981, Prendergast 1999). In contrast, our framework highlights that authority is never automatically established; its maintenance is a deliberate process that defines the political nature of organizations.[3] Shifting requirements for sustaining authority feed back into and dictate the level of internal competition, ultimately shaping incentive outcomes. By endogenizing this organizational constraint, our logic overcomes the limited leverage of traditional incentive theories in explaining the pronounced differences in organizational productivity and economic inequality observed across different institutional contexts.

Third, our paper advances the political selection literature on the "loyalty-competence trade-off" in governance, particularly within bureaucratic and autocratic settings (Egorov and Sonin 2011; Jia et al. 2015, Zakharov 2016, Dessein and Garicano 2025). Existing models typically treat the tension between political alignment and administrative merit as an exogenous structural dilemma. We endogenize this trade-off by demonstrating

[2] This insight echoes a long tradition in political theory, from Hobbes's (1968) claim that common authority is a precondition for stable cooperation to Polanyi's (1944) view that economic activity is embedded in social and political institutions rather than functioning as an autonomous market process. All productive systems rest on authority: some actors issue directives and allocate resources, while others comply.

[3] This redefines favoritism (Prendergast and Topel 1996) not as an exogenous agency cost, but as an endogenously rational strategy for securing compliance.

that it is conditioned on the structure and scale of the leader's informal loyalty network. When the network is sufficiently large, the trade-off evaporates, enabling high-powered competition to persist without risking the leader's control. Crucially, whereas previous political selection models are often customized to specific regime types, our organizational framework cuts across conventional regime labels to demystify productivity divergence across both democracies and autocracies.

The paper proceeds as follows. Section 2 presents the baseline model. Section 3 discusses its practical implications. Section 4 extends the model to internalize institutional change. Section 5 discusses implications for institutional reforms, and Section 6 concludes.

The Model

Set Up

Organizational Structure and Agent Types. Consider an organization with a three-tier hierarchical structure. A superior S occupies the top tier, n subordinates populate the bottom tier, and a single position or reward $V > 0$ is available at the intermediate level. One subordinate may be promoted to the intermediate position or receive the reward V.

The superior specifies the rules of internal competition, under which subordinates exert costly effort to compete for V. Subordinates are identical in ability and effort costs, but they differ in their degree of loyalty to the superior. Based on this heterogeneity, subordinates are classified into two types: loyal (l) and disloyal (d). We assume that among the n subordinates, m are loyal and $n - m$ are disloyal, where $m \in \{0, 1, \ldots, n\}$.

Loyalty is defined in terms of post-promotion behavior. After receiving the position or reward V, a loyal subordinate provides voluntary compliance with the superior's directives, whereas a disloyal subordinate exhibits weaker compliance, increasing coordination frictions. Accordingly, the number of loyal subordinates m captures the superior's authority capital, with a larger m reflecting a higher level of existing political authority. We treat m as an exogenously given organizational characteristic reflecting pre-existing political authority.

Heterogeneous loyalty is captured by parameter $\alpha > 0$: promoting a loyal subordinate generates an additional payoff α for the superior, reflecting the benefit of voluntary compliance, while promoting a disloyal subordinate yields a payoff of $-\alpha$, reflecting the cost of reduced compliance. For any $\alpha > 0$, whenever both types are eligible for V, the superior strictly prefers a loyal subordinate to a disloyal one, denoted by

$$l \succ d.$$

Legal Strength. The symmetric authority payoff $\pm\alpha$ makes α a convenient index of legal strength. We interpret a smaller α as stronger legal enforcement: as contracts become more enforceable, contractual authority more effectively secures compliance, reducing both the marginal value of loyalty and the marginal cost of disloyalty. Accordingly, the payoff gap between loyal and disloyal subordinates (2α) shrinks, making the two types less distinct. Conversely, under weaker legal enforcement, superiors rely more heavily on personalized political authority, which widens the payoff gap and is reflected in a larger α. Hence, cross-institutional variation in α captures differences in legal strength.

Adjustable Competition Rule. Scarcity of promotions and rewards at higher hierarchical tiers makes internal competition inevitable. Unlike market competition, however, competition within organizations is institutionally designed and reflects managerial discretion. We capture managerial discretion through an eligibility threshold $k \in \{1, \ldots, n\}$ chosen by the superior, which determines how many top-ranked subordinates are eligible for promotion or rewards. By varying k, the superior adjusts the intensity of performance-based competition within the organization. The extreme case $k = 1$ corresponds to a strict tournament in which only the top-ranked subordinate is eligible, while $k = n$ eliminates meritocratic competitiveness altogether, making all subordinates eligible regardless of rank. Thus, by choosing k, the superior directly controls how tightly promotion or reward V are tied to relative performance. This assumption creates scope for the superior to strengthen political authority by promoting or rewarding loyal subordinates within the eligible set k subject to her own discretion.

Stochastic Favoritism. We assume that loyal subordinates are structurally favored over disloyal ones conditional on ranking within the top k performers. Favoritism therefore operates only within the eligible set and remains disciplined by performance thresholds. Since entry into the top k is itself uncertain, favoritism is inherently probabilistic rather than deterministic. For any eligible loyal subordinates, the final selection depends on the composition of the top-k group. If there are $q \in \{1, \ldots, k\}$ loyal subordinates among the top k performers, the superior selects one of them uniformly at random, so each loyal subordinate in the eligible set is promoted with conditional probability $1/q$.

Symmetrically, the disadvantage faced by disloyal subordinates is also probabilistic. A disloyal subordinate's entry into the top k is uncertain, and ranking within the eligible set does not automatically preclude promotion. A disloyal subordinate still obtains a strictly positive promotion probability when all top-k performers are disloyal, in which case each is selected with equal probability $1/k$. By contrast, if at least one loyal subordinate ranks within the top k, disloyal subordinates receive zero probability of promotion or reward V. Thus, disloyalty entails a stochastic disadvantage that depends on both relative performance and the composition of the eligible set.

The Superior. The superior cares not only about aggregate organizational output but also about maintaining sufficient authority for coordination. She chooses the competition intensity k to maximize her utility, given by

$$U_S \equiv \max_{k\in\{1,\dots,n\}} \left\{ \sum_{i=1}^{m} x_{l_i}(k) + \sum_{j=1}^{n-m} x_{d_j}(k) + \alpha\big(P_l(k) - P_d(k)\big) \right\}. \tag{1}$$

where $\sum_{i=1}^{m} x_{l_i} + \sum_{j=1}^{n-m} x_{d_j}$ denotes expected aggregate productivity, defined as the sum of effort levels exerted by loyal subordinates $\{x_{l_i}\}_{i=1}^{m}$ and disloyal subordinates $\{x_{d_j}\}_{j=1}^{n-m}$.[4] The terms $P_l(k)$ and $P_d(k)$ denote the probabilities that a loyal and a disloyal subordinate, respectively, prevails in internal competition under competition intensity k. The term $\alpha\big(P_l(k) - P_d(k)\big)$ represents the superior's expected political authority payoff generated by competition outcomes. It captures the net contribution of internal competition to political authority, as determined by the relative likelihood that loyal rather than disloyal subordinates get promotion or reward V.

The Subordinates. After observing the competition intensity k, loyal and disloyal subordinates choose their effort levels $\{x_{l_i}\}_{i=1}^{m}$ and $\{x_{d_j}\}_{j=1}^{n-m}$, respectively, to maximize their expected utilities. Taking k as given, the problem of a loyal subordinate $i \in \{1,\dots,m\}$ is given by

$$U_{l_i} \equiv \max_{x_{l_i}\geq 0} \left\{ P_{l_i}\big(x_{l_i}, x_{-i}; k\big)\, V - \frac{x_{l_i}^2}{2} \right\}. \tag{2}$$

where the term $P_{l_i}(x_{l_i}, x_{-i}; k)$ represents the probability that loyal subordinate i obtains reward/position in the middle layer under competition intensity k, which depends on his own effort x_{l_i}, the efforts of competing subordinates x_{-i}, the competition intensity k and the composition of the top-k group, in particular the number q of loyal subordinates among the top k performers. We assume that this probability is increasing in x_{l_i} and decreasing in the efforts of competing subordinates, as standard in contest models. $V > 0$ denotes the value of the reward/position in the middle layer. The quadratic term $\frac{x_{l_i}^2}{2}$ captures the convex cost of exerting effort. Thus, each loyal subordinate chooses effort to trade off the expected benefit from winning the internal competition against the cost of effort.

Taking k as given, the problem of a disloyal subordinate $j \in \{1,\dots,n-m\}$ is given

[4] We deliberately adopt a neutral aggregation of effort to isolate the incentive effects of authority maintenance from technological complementarities.

by

$$U_{d_j} \;\equiv\; \max_{x_{d_j} \geq 0} \left\{ P_{d_j}\left(x_{d_j}, x_{-j}; k; q\right) V - \frac{x_{d_j}^2}{2} \right\}. \tag{3}$$

where $P_{d_j}(x_{d_j}, x_{-j}; k)$ represents the probability that disloyal subordinate j obtains the promotion or reward V under competition intensity k. The interpretation of $P_{d_j}(\cdot)$ follows analogously. The quadratic term $\frac{x_{d_j}^2}{2}$ captures the convex cost of exerting effort.

Winning Probabilities

Under stochastic favoritism, the winning probabilities of loyal and disloyal subordinates are piecewise defined across two cases, depending on whether the number of disloyal subordinates is sufficient to fill the top-k group. When $n - m \geq k$, the number of disloyal subordinates $n - m$ is no less than k, so it is feasible that the entire top-k group consists of disloyal subordinates. Hence, the winning probability for disloyal subordinates is positive. When $n - m < k$, there are fewer than k disloyal subordinates in total, implying that the top-k group must contain at least one loyal subordinate in every realization. Therefore, disloyal subordinates are excluded from selection with certainty, and their promotion probability is zero. Thus, the winning probability for a loyal subordinate $i \in \{1, \ldots, m\}$ is given by:[5]

$$P_{l_i} = \begin{cases} \dfrac{x_{l_i} \frac{1}{m}\left[\binom{n}{k} - \binom{n-m}{k}\right] + x_{l_{\bar{i}}}\left[\frac{m-1}{m}\binom{n}{k} - \binom{n-1}{k} + \frac{1}{m}\binom{n-m}{k}\right] + (n-m)x_{d_j} \frac{1}{m}\left[\binom{n-1}{k-1} - \binom{n-m-1}{k-1}\right]}{\binom{n-1}{k-1}\left[x_{l_i} + (m-1)x_{l_{\bar{i}}} + (n-m)x_{d_j}\right]}, & \text{if } n - m \geq k, \\ \dfrac{\frac{1}{m}\binom{n}{k} x_{l_i} + (m-1)x_{l_{\bar{i}}}\left(\frac{1}{m}\binom{n}{k} - \frac{1}{m-1}\binom{n-1}{k} + \frac{1}{m(m-1)}\binom{n-m}{k}\right)}{\binom{n-1}{k-1}\left[x_{l_i} + (m-1)x_{l_{\bar{i}}}\right]}, & \text{if } n - m < k. \end{cases} \tag{4}$$

$$P_{l_i} = \begin{cases} \dfrac{\begin{array}{l} x_{l_i} \sum\limits_{q=1}^{\min(m,k)} \frac{1}{q}\binom{m-1}{q-1}\binom{n-m}{k-q} + (m-1)x_{l_{\bar{i}}} \sum\limits_{q=2}^{\min(m,k)} \frac{1}{q}\binom{m-2}{q-2}\binom{n-m}{k-q} \\ +(n-m)x_{d_j} \sum\limits_{q=1}^{\min(m,k)} \frac{1}{q}\binom{m-1}{q-1}\binom{n-m-1}{k-q-1} \end{array}}{\binom{n-1}{k-1}\left[x_{l_i} + (m-1)x_{l_{\bar{i}}} + (n-m)x_{d_j}\right]} & \text{if } n - m \geq k \\ \dfrac{\sum\limits_{j=0}^{n-m} \frac{1}{k-j}\binom{n-m}{j}\binom{m-1}{k-j-1}x_{l_i} + (m-1)\sum\limits_{j=0}^{n-m} \frac{1}{k-j}\binom{n-m}{j}\binom{m-2}{k-j-2}x_{l_{\bar{i}}}}{\binom{n-1}{k-1}\left[x_{l_i} + (m-1)x_{l_{\bar{i}}}\right]}, & \text{if } n - m < k \end{cases} \tag{5}$$

[5]The mathematical expressions of P_{l_i} and P_{d_j} are based on Berry's (1993) Contest Success Function that describes an agent's probability of being rewarded in a competition with multiple winners. To incorporate authority maintenance, we develop Berry's (1993) Contest Success Function by including the winning likelihood conditioning on the performance that has ranked in the top k. See Appendix for more details.

The winning probability for disloyal subordinate $j \in \{1, \ldots, n-m\}$ is given by:

$$P_{d_j} = \begin{cases} \dfrac{1}{k} \dfrac{\binom{n-m-1}{k-1} x_{d_j} + (n-m-1)\binom{n-m-2}{k-2} x_{d_{\bar{j}}}}{\binom{n-1}{k-1}\left(x_{d_j} + (n-m-1)x_{d_{\bar{j}}} + m x_{l_i}\right)}, & \text{if } n-m \geq k, \\ 0, & \text{if } n-m < k. \end{cases} \tag{6}$$

$$P_{d_j} = \begin{cases} \frac{1}{k} \frac{\binom{n-m-1}{k-1} x_{d_j} + (n-m-1)\binom{n-m-2}{k-2} x_{d_{\bar{j}}}}{\binom{n-1}{k-1}[x_{d_j} + (n-m-1)x_{d_{\bar{j}}} + m x_{l_i}]} & \text{if } n-m \geq k \\ 0, & \text{if } n-m < k \end{cases} \tag{7}$$

Accordingly, the likelihood P_l that the superior can promote or reward any loyal subordinate, equivalently, the likelihood $1 - P_d$ that no disloyal subordinates win is given by:

$$P_l = 1 - P_d = \begin{cases} 1 - \frac{(n-m)\binom{n-m-1}{k-1} x_{d_j}}{\binom{n-1}{k-1}[(n-m)x_{d_j} + m x_{l_i}]} & \text{if } n-m \geq k \\ 1, & \text{if } n-m < k \end{cases} \tag{8}$$

Incentives, Effort, and Aggregate Output

Taking the competition intensity k as given, this section analyzes subordinates' equilibrium effort incentives and the resulting aggregate output.

Subordinates' effort provision. When $n - m \geq k$, loyal subordinate i's FOC is given by:

$$\frac{\partial P_{l_i}}{\partial x_{l_i}} V - x_{l_i} = 0 \tag{9}$$

Disloyal subordinate j's FOC is given by:

$$\frac{\partial P_{d_j}}{\partial x_{d_j}} V - x_{d_j} = 0 \tag{10}$$

The subordinates' reaction curve is given by:

$$x_{l_i} = B x_{d_j}, \qquad (B \geq 1). \tag{11}$$

where

$$B = \frac{\Delta + \sqrt{\Delta^2 + 4\binom{n-m}{k}\left(\binom{n-1}{k} - \binom{n-m-1}{k}\right)}}{2 \cdot \dfrac{m}{n-m}\binom{n-m}{k}}, \qquad \Delta \equiv \binom{n-1}{k} - \binom{n-m}{k} - \binom{n-m-1}{k}.$$

The best-response correspondence of loyal and disloyal subordinates are given by

$$
(x_l^*, x_d^*) = \begin{cases} \left(B\sqrt{\frac{V\binom{n-m}{k}(mB+n-m-k)}{\binom{n-1}{k-1}(n-m)(n-m+mB)^2}}, \sqrt{\frac{V\binom{n-m}{k}(mB+n-m-k)}{\binom{n-1}{k-1}(n-m)(n-m+mB)^2}}\right), & \text{if } n-m \geq k, \\ \left(\dfrac{1}{m}\left[\dfrac{\sum_{j=0}^{n-m}\binom{n-m}{j}\binom{m-1}{k-j}}{\binom{n-1}{k-1}}V\right]^{\frac{1}{2}}, 0\right), & \text{if } n-m < k. \end{cases} \tag{13}
$$

Proofs are provided in the Appendix.

Lemma 1 (Effort Asymmetry under Stochastic Favoritism)**.** *In the symmetric equilibrium, loyal subordinates exert weakly higher effort than disloyal subordinates:*

$$x_l^* \geq x_d^*.$$

Lemma 1 characterizes effort incentives when internal competition remains operative. When $n - m \geq k > 1$, favoritism applies within the eligible set, so loyal subordinates enjoy a strictly higher marginal return to effort than disloyal ones. Anticipating priority in the final selection conditional on ranking within the top k, loyal subordinates invest more aggressively in effort provision. By contrast, disloyal subordinates internalize the risk that even conditional on entering the eligible set, their winning probability collapses to zero whenever a loyal subordinate is present, which depresses their effort incentives.

The boundary case $k = 1$ corresponds to a strict tournament. Since favoritism cannot operate when only the top-ranked performer is eligible, loyal and disloyal subordinates face identical winning probabilities. Accordingly, effort provision is symmetric in this case, yielding $x_l^* = x_d^*$.

Corollary 1 (Effort Collapse under Degenerate Competition)**.** *Suppose* $n - m < k$.

If $k < n$*, disloyal subordinates optimally exert zero effort, while loyal subordinates continue to exert positive effort due to residual competition among loyal agents.*

If $k = n$*, promotion becomes completely independent of relative performance and depends solely on subordinate types. In this case, internal competition collapses entirely and both loyal and disloyal subordinates optimally exert zero effort in equilibrium.*

When $n - m < k < n$, at least one loyal subordinate must rank within the eligible set in every realization. As a result, disloyal subordinates are excluded from winning with certainty and optimally choose zero effort. Loyal subordinates, by contrast, continue to exert positive effort as long as $k < n$, reflecting residual competition among loyal agents within the eligible set.

When $k = n$, eligibility is unrestricted and promotion is entirely decoupled from performance. Since winning probabilities no longer depend on effort, all subordinates optimally choose zero effort and incentive provision collapses.

Aggregate Output Using the Vandermonde identity, aggregate output in the organization is given by

$$E^*(k) = mx_l^* + (n-m)x_d^* = \sqrt{V\,\frac{n-k}{k}}. \tag{14}$$

Proposition 1 (Aggregate Output Invariance)**.** *For any given organizational size n and prize value V, aggregate effort in equilibrium is independent of political authority capital m and strictly decreasing in the competition intensity parameter k.*

Proof. Substituting the equilibrium effort ratio B into the expression for total effort and applying standard binomial identities yields

$$(E^*)^2 = \frac{V\binom{n-1}{k}}{\binom{n-1}{k-1}},$$

which is independent of m. The detailed derivation is provided in Appendix. □

An increase in k expands the eligible set, diluting individual winning probabilities and weakening effort incentives, thereby reducing aggregate effort. However, for a given competition intensity k, increasing the number of loyal subordinates m mechanically replaces disloyal subordinates with loyal ones. Although a larger m intensifies competition within the eligible set and reduces each loyal subordinate's individual winning probability, thus lowering individual effort incentives, this incentive dilution is exactly offset by a composition effect. Loyal subordinates exert higher equilibrium effort than disloyal ones, so replacing a disloyal subordinate with a loyal subordinate raises per-capita effort at the margin. As a result, the reduction in individual effort caused by intensified within-group competition is compensated by the higher effort level of the additional loyal subordinate. Consequently, while competition intensity k directly governs aggregate output, variations in authority capital m affect the distribution of effort across agents but do not alter total effort as long as k remains fixed.[6]

[6] It is important to emphasize that this invariance result holds only in partial equilibrium, where the competition intensity k is taken as given. In this environment, changes in authority capital m affect the composition of effort across loyal and disloyal subordinates but do not alter aggregate output.

In general equilibrium, however, the superior optimally adjusts the competition intensity k in response to changes in m. Authority capital thus influences aggregate output indirectly, by shaping the superior's choice of k, which in turn governs subordinates' effort incentives. Accordingly, while m does not affect total effort conditional on k, it plays a central role in determining aggregate output through its impact on the equilibrium competition rule.

Visual Illustration: Effort levels and Aggregate Output in Partial Equilibrium Figure 1 depicts the relationship between authority capital m and competition intensity k, which may be selected by the superior S. It also illustrates subordinates' effort levels x_{l_i}, x_{d_j} and the overall productivity $mx_{l_i} + (n-m)x_{d_j}$ in the partial equilibrium.

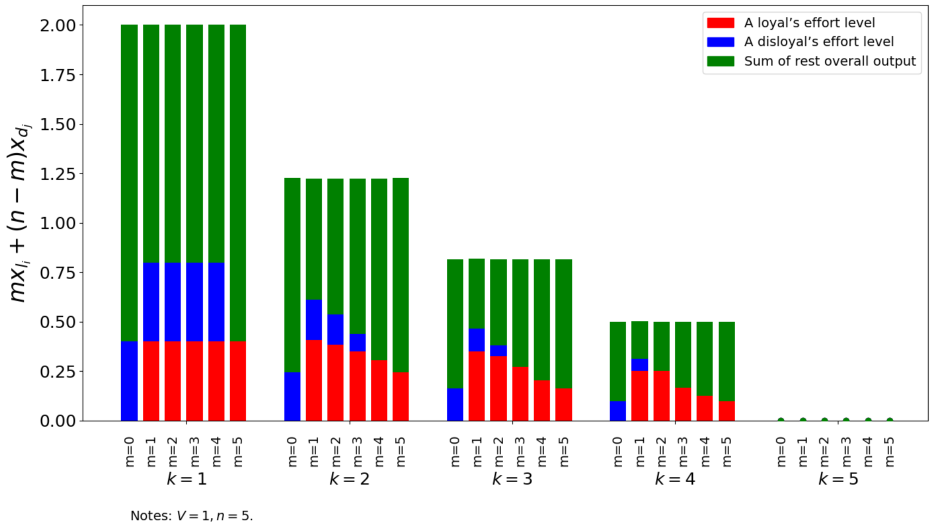


Figure 1 Effort levels and Aggregate Output in Partial Equilibrium

Consistent with the analytical results derived above, Figure 1 visually confirms that aggregate output is strictly decreasing in the competition intensity k, while remaining invariant to the level of authority capital m in the partial equilibrium.

Remark 1 (Resilience of loyal subordinates' effort)**.** *In partial equilibrium, effort provision by loyal subordinates is more resilient than that of disloyal subordinates to both weaker competition intensity (higher k) and greater authority capital m.*

Figure 1 reveals that although effort declines for both types as competition intensity weakens, the decline is markedly less pronounced for loyal subordinates. This asymmetry arises because loyalty preserves expected returns to effort even under diluted competition. As the eligible set expands, disloyal subordinates face an increasing risk that their winning probability collapses to zero whenever at least one loyal subordinate is present, sharply weakening their marginal incentives. By contrast, loyal subordinates retain priority within the eligible set, so their expected payoff from effort erodes more slowly as k increases.

A similar logic applies when authority capital m increases for a given k. While a larger m intensifies competition among loyal subordinates and reduces each loyal agent's individual winning probability, this effect is relatively moderate. For disloyal subordinates, an increase in m generates a double discouragement: it both reduces the likelihood that the top-k set can be filled entirely by disloyal agents and raises the probability that at least one loyal subordinate enters the eligible set, in which case disloyal agents' winning probability collapses to zero. As a result, effort incentives of disloyal subordinates deteriorate more sharply, reinforcing the greater resilience of loyal subordinates' effort provision.

Visual Illustration: Incentive Divergence and Effort Gaps in Partial Equilibrium. Figure 2 illustrates effort gaps $x_{l_i} - x_{d_j}$ between the loyal and disloyal subordinates for $n = 5$ and $V = 1$ across values of k and authority capital m.

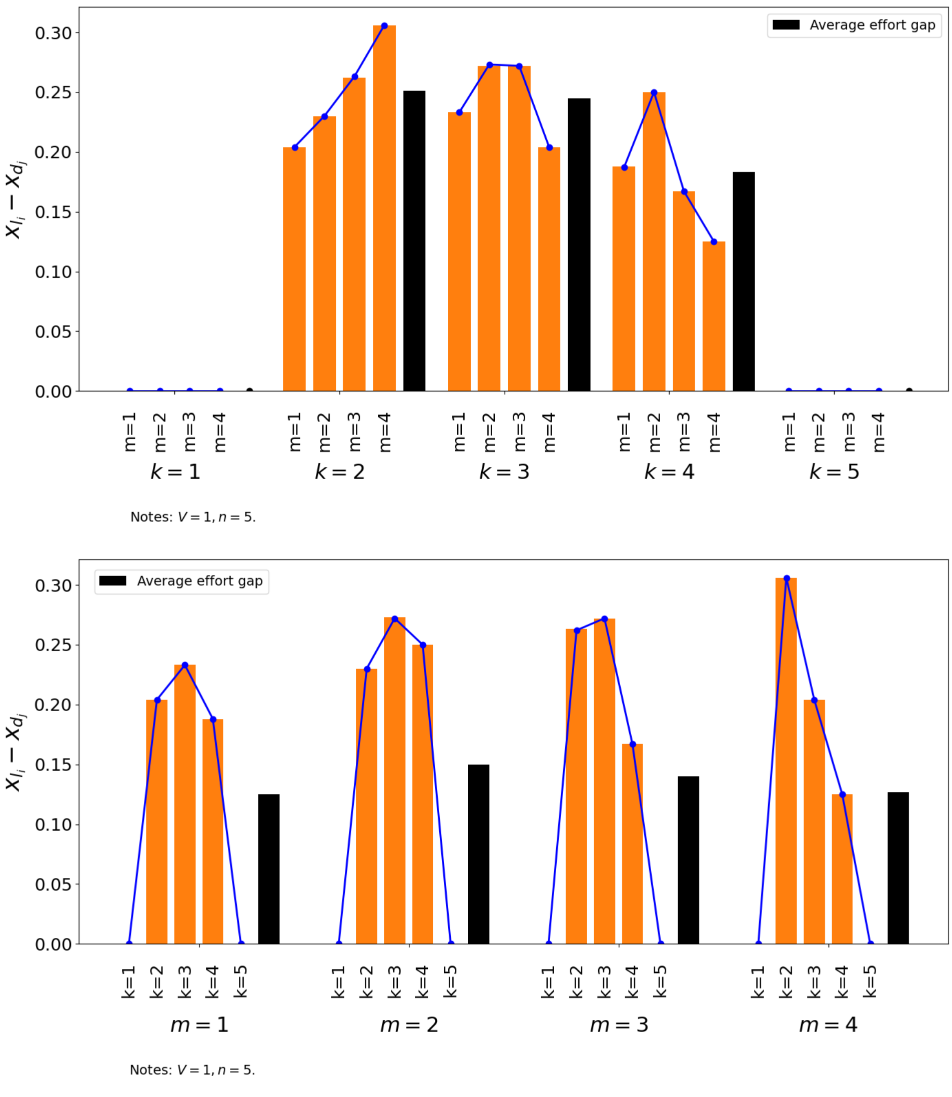


Figure 2 Effort gaps between the Loyal and Disloyal

Remark 2 (Hump-shaped effort gaps)**.** *In partial equilibrium, the effort gap between loyal and disloyal subordinates is hump-shaped in competition intensity k: it vanishes at the extremes and peaks at intermediate values.*

Figure 2 illustrates how favoritism creates a systematic wedge in effort provision. Disloyal subordinates are disproportionately discouraged because their winning probability collapses whenever at least one loyal subordinate enters the eligible set, whereas loyal subordinates retain priority conditional on eligibility.

As competition intensity weakens (higher k), effort incentives decline for both types, but disloyal effort falls more sharply. Consequently, the effort gap remains sizable over an intermediate range of k and collapses only when competition becomes either perfectly meritocratic ($k = 1$) or entirely non-meritocratic ($k = n$).

Holding k fixed, a larger authority capital m amplifies this asymmetry: increasing m simultaneously reduces the likelihood that the top-k set consists entirely of disloyal subordinates and raises the probability that at least one loyal subordinate appears, further depressing disloyal incentives relative to loyal ones.

This effort-gap pattern is a *partial-equilibrium* object. In the final equilibrium, m affects outcomes jointly with legal strength α through the superior's endogenous choice of competition intensity $k^*(m, \alpha)$, which determines both aggregate output and the distribution of effort gaps across institutional environments.

The Optimal Choice of Competition

The superior chooses competition intensity $k \in \{1, \ldots, n-1\}$ to maximize

$$U_S(k; \alpha, m) = g(k) + \alpha\big(1 - 2P_d(k; m)\big), \qquad g(k) = \sqrt{V\frac{n-k}{k}},$$

Since k is an integer choice, optimality is characterized by adjacent comparisons. Define the discrete payoff difference

$$\Delta U_S(k; \alpha, m) = U_S(k+1; \alpha, m) - U_S(k; \alpha, m) = \Delta g(k) - 2\alpha\, \Delta P_d(k; m), \tag{15}$$

where $\Delta g(k) = g(k+1) - g(k) < 0$ and $\Delta P_d(k; m) = P_d(k+1; m) - P_d(k; m) < 0$.

For each k, define the switching threshold

$$\alpha_k(m) = \frac{\Delta g(k)}{2\, \Delta P_d(k; m)} = \frac{g(k+1) - g(k)}{2\,[P_d(k+1; m) - P_d(k; m)]}, \tag{16}$$

which is well-defined and positive.

The sequence $\{\alpha_k(m)\}$ partitions the parameter space of α into intervals. In particular, the superior prefers $k+1$ to k if and only if $\alpha \geq \alpha_k(m)$. Consequently, the optimal competition intensity is characterized as

$$k^*(\alpha, m) = k \quad \Longleftrightarrow \quad \alpha \in \big[\alpha_{k-1}(m),\, \alpha_k(m)\big],$$

with the convention $\alpha_0 = 0$ and $\alpha_{n-1} = +\infty$.

Lemma 2 (Benchmark without Authority Concerns). *When $\alpha = 0$, the superior chooses the most intense competition rule $k^* = 1$, which yields the highest aggregate output.*

When $\alpha = 0$, the superior derives no authority payoff from promotion outcomes and therefore maximizes aggregate output alone. In this benchmark case, the superior's problem reduces to choosing the competition intensity k to maximize total effort. Since aggregate output is strictly decreasing in k, the optimal competition rule corresponds to a strict tournament, $k^* = 1$, which yields the first-best aggregate output.

Importantly, this benchmark does not imply that first-best competition cannot arise once authority concerns are introduced. As shown in the numerical analysis below, even when $\alpha > 0$, the superior may still optimally choose $k^* = 1$ over a nontrivial range of parameter values.

Proposition 2 (Optimal competition intensity). *For fixed n and V, the superior's optimal competition intensity $k^*(\alpha, m)$ satisfies the following comparative statics:*

1. *For a given legal environment α, $k^*(\alpha, m)$ is weakly decreasing in authority capital m.*

2. *For a given authority capital m, $k^*(\alpha, m)$ is weakly increasing in α.*

The intuition is as follows. A larger authority capital m raises the likelihood that loyal subordinates enter the eligible set, thereby increasing the superior's expected authority payoff from promotion outcomes. To preserve these authority rents, the superior optimally intensifies competition by choosing a smaller k.

By contrast, a larger α increases the relative importance of authority maintenance vis-à-vis aggregate output. To mitigate the risk of promoting disloyal subordinates, the superior optimally expands the eligible set and relaxes internal competition, resulting in a higher k^*.

Visual Illustration: Optimal Competition Intensity and Aggregate Output. Figure 3 illustrates Proposition 2 using a numerical example with $n = 5$ and $V = 1$. The color gradient in Figure 3 further illustrates the productivity implications implied by Proposition 1: darker cells correspond to lower k^* and therefore higher aggregate output.

The figure highlights three patterns. First, for a fixed authority capital m, the optimal competition intensity k^* increases monotonically with α, reflecting the superior's growing emphasis on authority maintenance under weaker legal enforcement. Second, for a given α, higher authority capital m induces more intense competition, manifested by a lower k^*. Third, the benchmark case $\alpha = 0$ yields $k^* = 1$ for all m, consistent with the first-best strict tournament outcome.

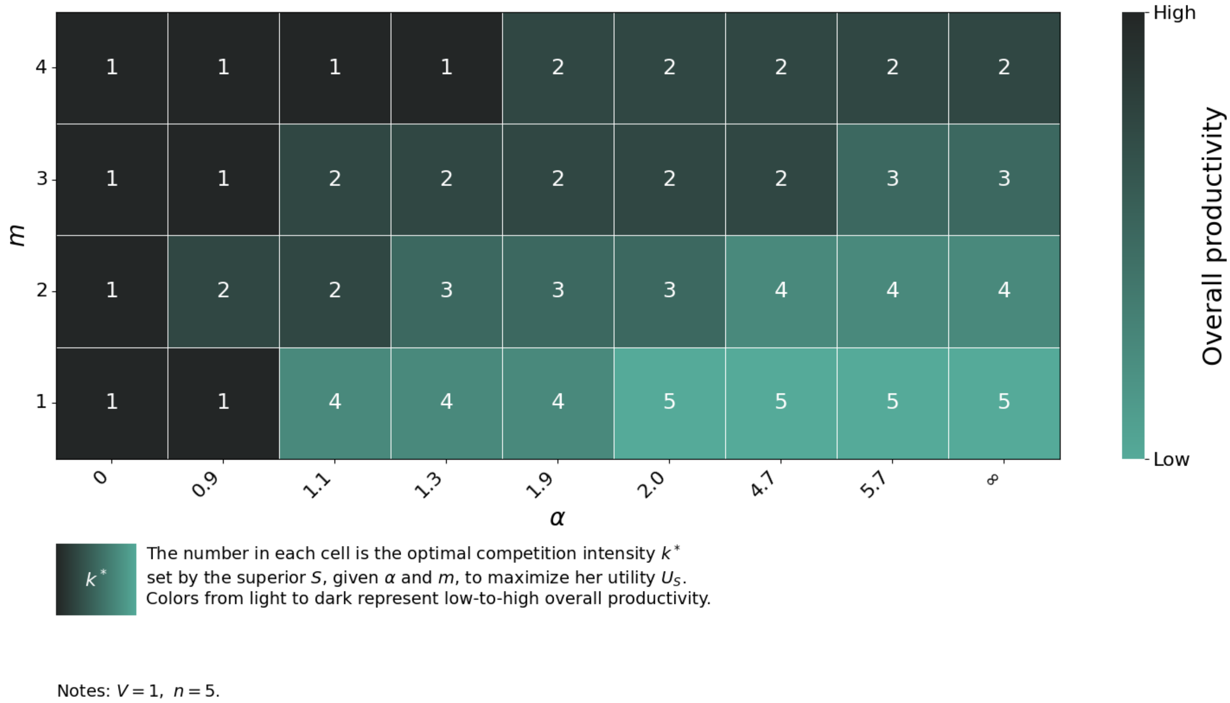


Figure 3 OPTIMAL COMPETITION INTENSITY ACROSS AUTHORITY CAPITAL AND LEGAL STRENGTH

Corollary 2 (Induced Productivity Effects)**.** *In the final equilibrium, aggregate organizational output varies systematically with authority capital m and legal strength α through the superior's endogenous choice of competition intensity $k^*(\alpha, m)$.*

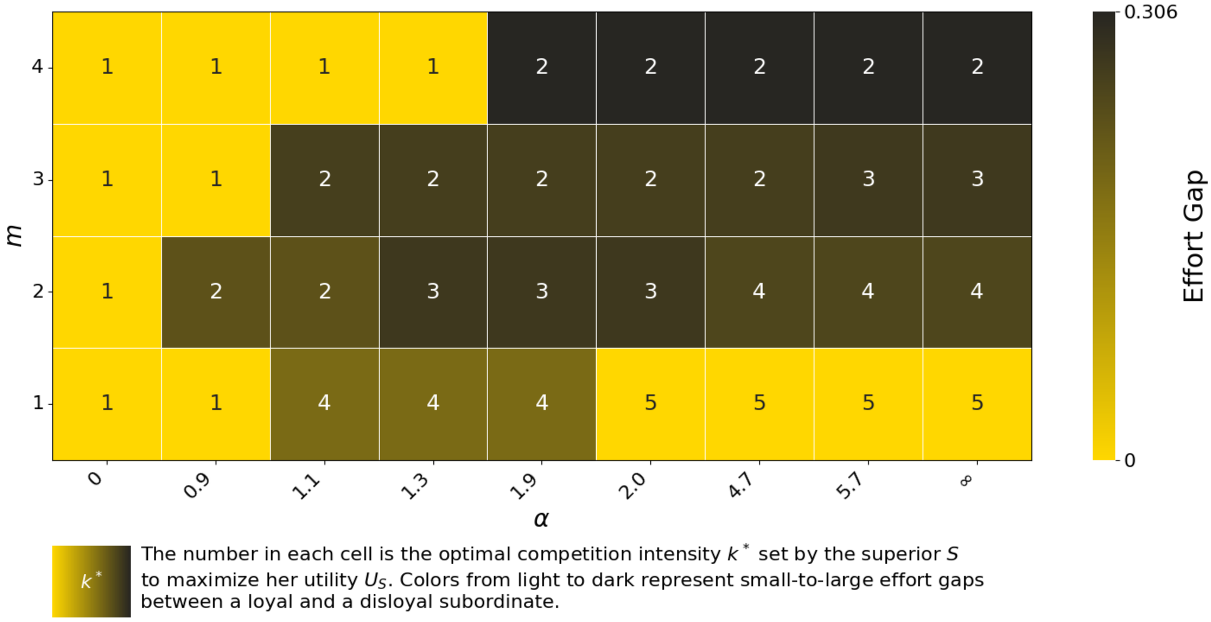
Specifically, environments in which authority capital or legal enforcement induces the superior to choose more intense competition (lower k^*) sustain higher aggregate output, whereas weaker legal environments that lead to relaxed competition (higher k^*) exhibit lower productivity.

This result follows directly from the fact that aggregate output is strictly decreasing in k in partial equilibrium.

Visual Illustration: Optimal Competition Intensity and Equilibrium Effort Gaps. Figure 4 depicts the relationship between m, k^*, α, and the effort gaps between the loyal and disloyal $x_{l_i} - x_{d_j}$ when $V = 1$ and $n = 5$.

As shown in Figure 4, effort gaps emerge only when the superior chooses an intermediate competition rule, $1 < k^* < n$. When competition is maximally intense ($k^* = 1$), favoritism cannot operate and loyal and disloyal subordinates exert identical effort. When competition is fully relaxed ($k^* = n$), promotion outcomes depend solely on identity rather than performance, leading both types to optimally slack off and eliminating effort differences.

Proposition 3 (Political origin of effort gaps)**.** *Effort gaps are political in origin:*



Notes: $V = 1,\ n = 5$.

Figure 4 Competition Intensity, Authority Capital and Effort Gaps

they can arise from authority maintenance within organizations rather than from market-based differences in productivity or skills.

Specifically, effort gaps arise when the superior optimally chooses an intermediate competition rule that allows loyalty-based prioritization to operate. Authority capital m and legal strength α shape the magnitude and incidence of effort inequality only indirectly, through their effect on the endogenous choice of competition intensity.

Therefore, effort inequality in equilibrium is not a monotone function of institutional strength, but a political outcome generated by authority-preserving competition design.

Remark 3 (Effort Gaps and Productivity Are Not Monotonically Linked)**.** *Figure 4 highlights that equilibrium effort gaps and aggregate productivity need not move together. High productivity can coexist with minimal effort gaps (e.g., under strict tournaments with $k^* = 1$), but also with substantial effort gaps under intermediate competition rules. Conversely, low productivity may arise both in environments with negligible effort gaps ($k^* = n$) and in settings with pronounced gaps.*

This absence of a monotonic relationship reflects the fact that productivity is governed primarily by the intensity of competition, whereas effort gaps depend on how favoritism interacts with competition through authority capital and legal enforcement. As a result, similar levels of productivity can be associated with very different distributions of effort across subordinate types.

Practical Implication

The model yields several practical implications for understanding productivity, inequality and the political nature of hierarchical organizations, ranging from firms to states.

A central implication is that differences in productivity need not originate from differences in technology, culture, or regime type per se, but can arise endogenously from how internal competition is designed under authority constraints. When superiors rely heavily on loyalty-based authority, they may strategically adjust competition intensity to preserve authority rents, even at the cost of weaker incentives for effort. As a result, otherwise similar organizations may exhibit sharply different productivity outcomes simply because authority capital and legal enforcement shape different competition rules.

This perspective helps reconcile long-standing disagreements in the literature on political regimes and economic performance. Rather than attributing growth differences to democracy or autocracy as such, the model suggests that both democratic and non-democratic systems can sustain either high or low productivity, depending on how authority concerns interact with internal competition. Weak legal environments do not mechanically imply low productivity: when authority capital is sufficiently strong, superiors may still enforce intense internal competition, sustaining high levels of effort and output. Conversely, even formally democratic systems may experience declining productivity if authority concerns lead to relaxed competition.

The same logic applies at the organizational level. Firms with similar technologies may display large productivity gaps if leaders differ in their authority capital or reliance on loyalty-based authority (legal strength) and therefore choose different competition rules for promotion and advancement. In this sense, competition intensity emerges as an institutional outcome shaped by authority considerations rather than a purely technological or market-driven parameter.

The model also offers a new interpretation of economic inequality within organizations. Effort gaps between favored and disfavored subordinates do not arise as a direct consequence of low productivity or weak competition. Instead, they are a political byproduct of authority-preserving competition design. Severe effort inequality may coexist with high productivity under intermediate competition rules, while minimal inequality can accompany both the most intense and the weakest forms of competition. This helps explain why economic inequality is observed across a wide range of organizational and institutional environments without a stable monotonic relationship to aggregate performance. While the model is developed in an organizational setting, its logic naturally extends to state-level governance. States, like large hierarchical organizations, rely on authority rather than prices to coordinate behavior. When political authority is maintained through identity-based favoritism, incentive structures become uneven across groups. As a result, inequality emerges not primarily from differential market returns, but from systematically induced differences in effort and participation. Economic inequality, in this view, is a downstream consequence of politically generated effort inequality. This implication complements political science theories of social cleavages by identifying an incentive-based mechanism through which identity divisions translate into persistent economic inequality (Lipset and Rokkan 1967; Alesina and Glaeser 2004; Padgett and McLean 2006).

Taken together, these implications highlight that productivity and inequality are jointly shaped—but not mechanically linked—by how authority is maintained through internal competition. Institutional reforms that focus solely on strengthening formal rules or increasing competition may therefore have limited effects unless they also account for the authority structures that govern how competition is implemented in practice.

Model Extension: Endogenous Authority Capital and Tipping Points

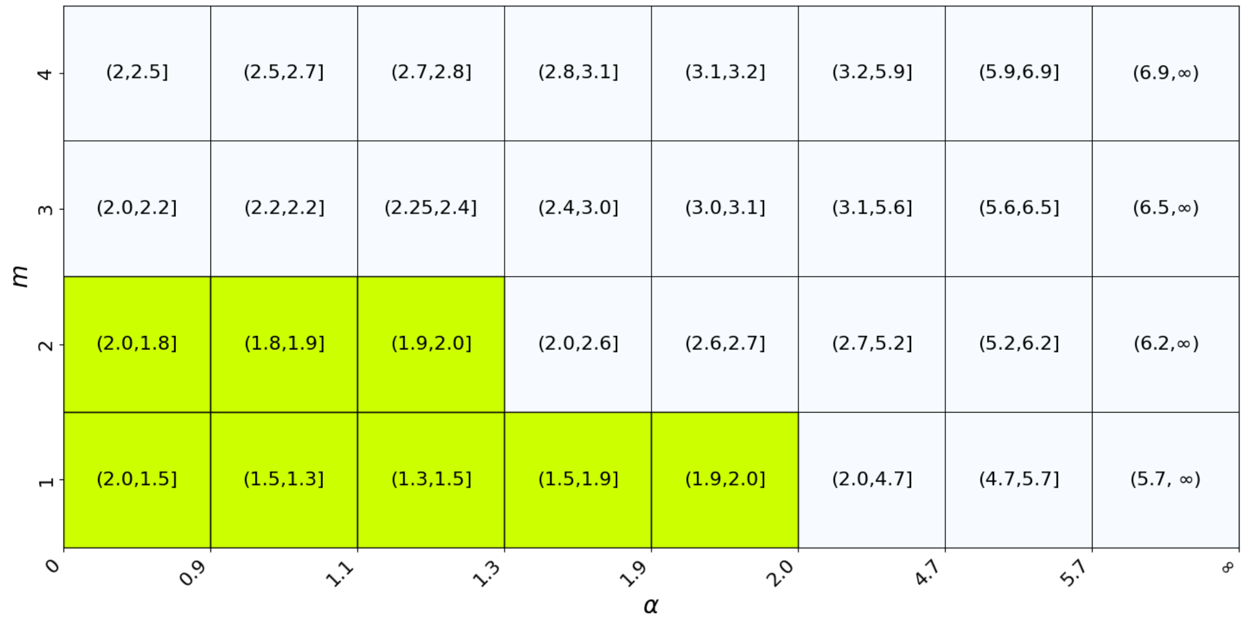
Our baseline analysis treats the choice of competition intensity k^* as the sole strategic margin available to the superior S, taking authority capital m and legal strength α as exogenously given. This formulation captures environments in which organizational and institutional constraints are slow-moving.

In practice, however, authority capital and legal strength are often shaped, at least in part, by strategic actions of the superior (Zingales 2017). Through organizational restructuring, selective promotion, coalition-building, or political contestation, a superior may expand or contract the pool of loyal subordinates, thereby influencing authority capital m. Similarly, legal strength α may be indirectly affected by discretionary enforcement, institutional reform, or deliberate weakening of formal constraints.

This section extends the model by allowing the superior to exercise limited discretion over authority capital and legal strength, and examines how these additional margins interact with competition design and equilibrium outcomes.

Figure 5 illustrates the superior's maximized utility U_S^* across combinations of authority capital m, legal strength α, and the induced optimal competition intensity k^* in a numerical example with $n = 5$ and $V = 1$. We first examine the superior's preference over authority capital m for a given α, and then turn to the choice of α conditional on m.

Proposition 4 (Authority Capital Expansion)**.** *Holding legal strength α fixed, the superior's maximized utility U_S^* is weakly increasing in authority capital m. As a result, expanding the pool of loyal subordinates is a dominant strategy whenever feasible.*



Notes: Each cell reports the interval of the superior S's maximized utility for the corresponding values of α and m.

Notes: $V = 1, n = 5$.

Figure 5 Superior's Maximized Utility Ranges

The intuition is as follows. An increase in authority capital m raises the likelihood that loyal subordinates enter the eligible set, thereby improving the promotion probability structure in favor of authority reinforcement, even if the superior does not adjust the competition rule.

This structural shift expands the superior's payoff-relevant choice set. With a larger pool of loyal subordinates, the superior can intensify competition by selecting a smaller k without sacrificing authority payoffs. In fact, stricter competition simultaneously raises aggregate output while preserving—or even enhancing—the probability that promotion outcomes reinforce authority.

As a result, an increase in authority capital unambiguously raises the superior's maximized utility. This generates a fundamental incentive for superiors to expand authority capital whenever feasible, reflecting an intrinsic tendency of hierarchical organizations to accumulate political authority.

Remark 4 (Tipping Points in Legal Strength)**.** *Figure 5 reveals that the relationship between legal strength α and the superior's maximized utility U_S^* is non-monotonic when authority capital m is small. For sufficiently low levels of α, marginal increases in authority reliance reduce the superior's payoff. Beyond a threshold value $\alpha_{tp}(m)$, further increases in α strictly raise U_S^*.*

When authority capital is small, the probability of rewarding a loyal subordinate is limited. In this region, marginal increases in α amplify authority concerns without generating sufficient authority rents, leading to a decline in U_S^*. Once authority reliance exceeds a tipping point α_{tp}, the superior optimally relaxes competition by increasing k^*, which raises the probability of rewarding loyal subordinates and turns expected authority gains positive.

Remark 5 (Disappearance of Tipping Points)**.** *As authority capital m increases, the tipping point $\alpha_{tp}(m)$ decreases and eventually disappears. For sufficiently large m, the superior's maximized utility U_S^* becomes monotonically increasing in α.*

A larger authority capital increases the baseline likelihood of rewarding loyal subordinates. As a result, even small increases in authority reliance (weaker legal strength) generate positive expected authority rents, making intensified competition less sustainable. Consequently, the superior's preference for strong legal strength and strict competition vanishes as m grows.

Implications: Authority Accumulation, Institutional Stability and Reform

The extension yields two central implications regarding the internal logic of hierarchical organizations and the long-run stability of institutional arrangements.

First, hierarchical organizations endogenously generate incentives for political authority accumulation—not as a distortion, but as a consequence of probability-based competition design. Whenever the superior is able to influence the composition of loyal subordinates, she strictly prefers to expand authority capital. This preference is not incidental, but rooted in the organizational structure that underpins the political nature of hierarchical organizations. Authority accumulation emerges as an endogenous adaptive response that allows organizations to maintain coordination and operational efficiency even under volatile or weak legal environments.

Crucially, this preference is robust across legal regimes. Increasing authority capital not only strengthens political control, but also expands the feasible set of internal competition designs that can sustain high organizational performance across heterogeneous levels of legal enforcement. Political authority accumulation is therefore not traded off against efficiency; rather, it enlarges the institutional space in which authority and productivity remain compatible.

This mechanism applies broadly—from firms and public bureaucracies to nation-states—wherever hierarchical structure and internal competition coexist.

Second, the model highlights why high levels of legal enforcement are difficult to sustain unless institutional reforms are sufficiently comprehensive. When legal strength is sufficiently strong and authority capital is small, marginal decline in legal strength may reduce the superior's utility, creating a region in which weakening legal strength is temporarily unattractive. However, this region is bounded by a tipping point. Once authority capital expands beyond a threshold, the superior's incentives reverse: weakening legal strength becomes privately optimal, even though high legal enforcement remains socially efficient.

This result implies that partial legal reforms may be inherently fragile. If reforms fail to push legal strength sufficiently far below the tipping point—or fail to constrain the superior's ability to expand authority capital—then the institutional environment cannot sustain itself endogenously. Over time, private incentives reassert themselves, and legal enforcement erodes.

Third, these dynamics provide a structural explanation for persistent institutional backsliding. Even when high legal enforcement and low inequality are initially achieved, they are not self-enforcing equilibria if superiors retain discretion over authority accumulation. Once the system crosses the tipping point, the superior has strong incentives to weaken legal constraints and expand political authority, despite full awareness that strong rule of law is socially beneficial.

From this perspective, democratic backsliding and rule-of-law erosion need not reflect cultural regression or technological stagnation. Instead, they may arise endogenously from the interaction between hierarchical authority, discretionary power, and probability-based competition. Sustaining a stable, high-rule-of-law equilibrium therefore requires institutions that simultaneously limit both political authority expansion and discretionary erosion of legal enforcement.

Conclusion

A central premise of organizational economics is that organizations allocate resources through authority rather than prices. Existing theories, however, have largely equated such authority with contractual authority—authority enforced by formal contracts and third-party legal enforcement. This abstraction obscures a fundamental feature of real organizations: contractual enforcement varies widely across countries and institutional environments, and is often incomplete even within advanced legal systems. When contractual authority is weak or imperfectly enforced, organizations face a basic problem—how to sustain sufficient internal authority to coordinate activities and ensure compliance.

This paper argues that organizations respond to this problem by endogenously maintaining political authority within their hierarchies. Political authority operates through personnel selection and promotion, where loyalty and obedience become relevant criteria alongside performance. By adjusting internal competition intensity and shaping who is hired, promoted, and rewarded, superiors can preserve effective authority even when formal enforcement is limited.

We show that the productivity consequences of authority maintenance are inherently institutional-contingent. When legal enforcement is strong, contractual authority is reliable, and superiors have little need to rely on political authority. In such environments, promotion decisions place greater weight on performance rather than loyalty, leading to more intense internal competition and higher aggregate output.

When legal strength is weak, contractual authority becomes fragile, and superiors must rely more heavily on political authority to maintain basic organizational control. This reliance increases the incentive to favor loyal subordinates and to relax internal competition by expanding the eligibility threshold. Importantly, weak legal enforcement is not, by itself, sufficient to generate low productivity. When authority capital is sufficiently strong—because a large pool of loyal subordinates already exists—superiors can maintain strict competition while still securing authority, thereby sustaining high organizational output even under weak legal institutions.

These results reconcile two seemingly conflicting observations. They are consistent with the conventional view that strong legal institutions promote economic efficiency, but they also explain why high productivity can persist in weak legal environments. In such cases, political authority substitutes for contractual authority, allowing organizations to preserve both control and performance through endogenous competition design.

Our model extension shows that authority accumulation is intrinsic to hierarchical organization rather than a distortion. Instead, it constitutes a core mechanism through which organizations maintain stable operation and sustain high productivity, even under weak or volatile legal enforcement—an aspect largely overlooked in the existing literature.

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Appendix

Theories of Winning Likelihood

The mathematical expression of the winning likelihood applied in this paper is based on Berry's (1993) Multi-winner Success Function. To understand Berry's (1993) Multi-winner Success Function and the extension of the winning likelihood function applied in our paper, it is necessary to introduce the Contest Success Function (CSF) and the Tullock Success Function (TSF) as the baseline.

The CSF is a core component of a contest situation, which depicts each competitor's winning likelihood given others' level of effort. Let $x = (x_1, x_2, ..., x_n)$ denote a vector of efforts for n players. Each player i's winning probability is denoted by $Prob^i(x)$ ($Prob^i(x) : [0, X]^n \to \mathbb{R}$ where $X > x_i$ for all $i \in N$). For a contest that only has one winner, each player's winning probability should meet the following three properties (Skaperdas 1996):

(1A)$\sum_{i \in N} Prob^i(x) = 1$ and $Prob^i(x) \geq 0$ for all $i \in N$ and all x; if $x_i > 0$ then $Prob^i(x) > 0$.

(2A) For all $i \in N$, $Prob^i(x)$ is increasing in x_i and decreasing in x_j for all $j \neq i$.

(3A) For any permutation π of N (i.e., a bijection $\pi : N \to N$), it has

$$Prob^{\pi(i)}(x) = Prob(x_{\pi_1}, x_{\pi_2}, ..., x_{\pi_n}) \forall i \in N.$$

Tullock Success Function (TSF) is expressed below,

$$Prob^i(x) = \frac{x_i^r}{x_1^r + x_2^r + ... + x_n^r} \tag{17}$$

TSF is a special type of CSF that meets the above three properties. It depicts the winning likelihood for an individual competitor in a single-winner contest. In the mathematical expression of TSF, the nominator only includes information relating to competitor i's effort level x_i because competitor i is the only winner in the contest. The denominator $\sum_{i=1}^{n} x_i^r$ is the sum of all information about each competitor's effort levels, and r refers to an index of external uncertainty/ the marginal cost of influencing the probability of winning (Rai and Sarin 2009). In a single-winner contest setting, the ratio of nominator and denominator of TSF is the winning likelihood for competitor i, given other competitors' effort level $x_{\bar{i}}$ and the uncertainty level r.

TSF functions as the baseline for Barry's Multi-winner Success Function. In a contest with k winners ($1 < k \leq n$), each player's winning probability would be more significant than when there is one winner. Hence, to describe the winning probability for each competitor in a multi-winner contest, Berry (1993) made some adjustments to

TSF while keeping the three properties of CFS.

Let $Prob_k^i(x)(i = 1, 2, ...n)$ denote competitor i's probability to get into the top k. In other words, $Prob_k^i(x)(i = 1, 2, ...n)$ depicts competitor i's winning likelihood in a k-winner contest. We have the expression of $Prob_k^i(x)$ below.

The numerator displays all possible combinations of winner i and the other $k-1$ winners, which are denoted as $j \neq i$. The expression in each bracket $(x^r_{j^1_q} + x^r_{j^2_q} + ... + x^r_{j^{k-1}_q})$, where $q = 1, ..., \binom{n-1}{k-1}$ represents the qth set of different combinations of the other winner j 's efforts.

In a k-winner contest of n players, there are $\binom{n-1}{k-1}$ possibilities for player i ($i = 1, ..., n$) to get into top k qualified, where $\binom{n-1}{k-1}$ represents the combination of $n-1$ objects taking $k-1$ at a time. $(x^r_{z^1_p} + x^r_{z^2_p} + ... + x^r_{z^k_p})$, $p = 1, ..., \binom{n}{k}$ in the denominator denotes the pth combination of any players that is possible to win the contest. Similarly, there are $\binom{n-1}{k-1}$ winning combinations in total. The ratio between the numerator and the denominator is calculated as the winning probability for competitor i in a multi (k)-winner competition.

$$
\begin{aligned}
Prob_k^i(x) &= \frac{\begin{array}{c} x_i^r + (x^r_{j^1_1} + x^r_{j^2_1} + ... + x^r_{j^{k-1}_1}) + x_i^r + (x^r_{j^1_2} + x^r_{j^2_2} + ... + x^r_{j^{k-1}_2}) \\ +... + x_i^r + \left(x^r_{j^1_{\binom{n-1}{k-1}}} + x^r_{j^2_{\binom{n-1}{k-1}}} + ... + x^r_{j^{k-1}_{\binom{n-1}{k-1}}} \right) \end{array}}{\begin{array}{c} (x^r_{z^1_1} + x^r_{z^2_1} + ... + x^r_{z^k_1}) + (x^r_{z^1_2} + x^r_{z^2_2} + ... + x^r_{z^k_2}) \\ +... + \left(x^r_{z^1_{\binom{n}{k}}} + x^r_{z^2_{\binom{n}{k}}} + ... + x^r_{z^k_{\binom{n}{k}}} \right) \end{array}} \\
&= \frac{\binom{n-1}{k-1} x_i^r + \binom{n-2}{k-2} \sum_{j \neq i} x_j^r}{\binom{n-1}{k-1} \sum_{z=1}^{n} x_z^r}
\end{aligned}
$$

Where $j \neq i$, $j, i \in N$.

Calculation Details for Section III

When $n - m \geq k$,

$$
\frac{\partial P_{l_i}}{\partial x_{l_i}} = \frac{\begin{gathered}\sum_{i=1}^{\min(m,k)} \frac{1}{i}\binom{m-1}{i-1}\binom{n-m}{k-i}[x_{l_i} + (m-1)x_{l_i} + (n-m)x_{d_j}] \\ - \left[x_{l_i} \sum_{i=1}^{\min(m,k)} \frac{1}{i}\binom{m-1}{i-1}\binom{n-m}{k-i} + (m-1)x_{l_i} \sum_{i=2}^{\min(m,k)} \frac{1}{i}\binom{m-2}{i-2}\binom{n-m}{k-i} \right. \\ \left. +(n-m)x_{d_j} \sum_{i=1}^{\min(m,k)} \frac{1}{i}\binom{m-1}{i-1}\binom{n-m-1}{k-i-1}\right]\end{gathered}}{\binom{n-1}{k-1}[x_{l_i} + (m-1)x_{l_i} + (n-m)x_{d_j}]^2}
$$

$$
= \frac{(\binom{n-m}{k-1} + \sum_{i=2}^{\min(m,k)} \frac{1}{i}\binom{m-2}{i-1}\binom{n-m}{k-i})(m-1)x_{l_i} + (n-m)x_{d_j} \sum_{i=1}^{\min(m,k)} \binom{m-1}{i-1}\binom{n-m-1}{k-i}}{\binom{n-1}{k-1}[x_{l_i} + (m-1)x_{l_i} + (n-m)x_{d_j}]^2}
$$

$$
= \frac{x_{l_i} \sum_{i=1}^{min(m,k)} \binom{m-1}{i}\binom{n-m}{k-i} + \frac{n-m}{m} x_{d_j} \sum_{i=1}^{min(m,k)} \binom{m}{i}\binom{n-m-1}{k-i}}{\binom{n-1}{k-1}[m x_{l_i+(n-m)x_{d_j}}]^2} \tag{19}
$$

When $n - m < k$, we get:

$$
\begin{aligned}
\frac{\partial P_{l_i}}{\partial x_{l_i}} &= \frac{(m-1)x_{l_i}(\sum_{i=0}^{n-m} \frac{1}{k-i}\binom{n-m}{i}\binom{m-1}{k-i-1} - \sum_{i=0}^{n-m} \frac{1}{k-i}\binom{n-m}{i}\binom{m-2}{k-i-2})}{\binom{n-1}{k-1}[x_{l_i} + (m-1)x_{l_i}]^2} \\
&= \frac{(m-1)x_{l_i} \sum_{i=0}^{n-m} \frac{1}{k-i}\binom{n-m}{i}\binom{m-2}{k-i-1}}{\binom{n-1}{k-1}[x_{l_i} + (m-1)x_{l_i}]^2} \\
&= \frac{\sum_{i=0}^{n-m} \binom{n-m}{i}\binom{m-1}{k-i}}{\binom{n-1}{k-1} m^2 x_{l_i}}
\end{aligned} \tag{20}
$$

When $n - m \geq k$, the calculation of $\frac{\partial P_{d_j}}{\partial x_{d_j}}$ is:

$$
\begin{aligned}
\frac{\partial P_{d_j}}{\partial x_{d_j}} &= \frac{\binom{n-m-1}{k-1}[x_{d_j}+(n-m-1)x_{d_j}+mx_{l_i}]-[\binom{n-m-1}{k-1}x_{d_j}+\binom{n-m-2}{k-2}(n-m-1)x_{d_j}]}{k\binom{n-1}{k-1}[x_{d_j}+(n-m-1)x_{d_j}+mx_{l_i}]^2}\\
&= \frac{\binom{n-m-1}{k-1}[mx_{l_i}+(n-m)x_{d_j}]-[\binom{n-m-1}{k-1}x_{d_j}]+\binom{n-m-2}{k-2}(n-m-1)x_{d_j}}{k\binom{n-1}{k-1}[mx_{l_i}+(n-m)x_{d_j}]^2}\\
&= \frac{\binom{n-m-1}{k-1}mx_{l_i}+(n-m-1)\binom{n-m-2}{k-1}x_{d_j}}{k\binom{n-1}{k-1}[mx_{l_i}+(n-m)x_{d_j}]^2}\\
&= \frac{km\binom{n-m}{k}x_{l_i}+k\binom{n-m-1}{k}x_{d_j}}{k(n-m)\binom{n-1}{k-1}[mx_{l_i}+(n-m)x_{d_j}]^2}\\
&= \frac{\frac{m}{n-m}\binom{n-m}{k}x_{l_i}+\binom{n-m-1}{k}x_{d_j}}{\binom{n-1}{k-1}[mx_{l_i}+(n-m)x_{d_j}]^2}
\end{aligned}
\tag{21}
$$

By combining equation (9) and equation (10), we get the expression of x_{l_i} as a function of x_{d_j} when $n-m \geq k$ as follows:

$$
\begin{aligned}
&\frac{x_{l_i}\sum\limits_{i=1}^{\min(m,k)}\binom{m-1}{i}\binom{n-m}{k-i}+\frac{n-m}{m}x_{d_j}\sum\limits_{i=1}^{\min(m,k)}\binom{m}{i}\binom{n-m-1}{k-i}}{x_{l_i}} = \frac{\binom{n-1}{k-1}[mx_{l_i}+(n-m)x_{d_j}]^2}{V}\\
&\Rightarrow \frac{mx_{l_i}\sum\limits_{i=1}^{\min(m,k)}\binom{m-1}{i}\binom{n-m}{k-i}+(n-m)x_{d_j}\sum\limits_{i=1}^{\min(m,k)}\binom{m}{i}\binom{n-m-1}{k-i}}{mx_{l_i}}\\
&\qquad = \frac{m\binom{n-m}{k}x_{l_i}+(n-m)\binom{n-m-1}{k}x_{d_j}}{(n-m)x_{d_j}}\\
&\Rightarrow x_{l_i} = \frac{\sum\limits_{i=1}^{\min(m,k)}\binom{m-1}{i}\binom{n-m}{k-i}-\binom{n-m-1}{k} + \left[(\sum\limits_{i=1}^{\min(m,k)}\binom{m-1}{i}\binom{n-m}{k-i}-\binom{n-m-1}{k})^2+4\binom{n-m}{k}\sum\limits_{i=1}^{\min(m,k)}\binom{m}{i}\binom{n-m-1}{k-i}\right]^{\frac{1}{2}}}{\frac{2m}{n-m}\binom{n-m}{k}}x_{d_j}
\end{aligned}
\tag{22}
$$

Table 1 PARTIAL EQUILIBRIUM EXAMPLE WITH $n = 5$

E / x_{l_i} / x_{d_j}		k = 1	2	3	4	5
m	0	2 N/A 0.400	1.225 N/A 0.245	0.816 N/A 0.163	0.500 N/A 0.100	0 N/A 0
	1	2 0.400 0.400	1.225 0.408 0.204	0.816 0.350 0.117	0.500 0.250 0.063	0 0 0
	2	2 0.400 0.400	1.225 0.383 0.153	0.816 0.327 0.054	0.500 0.250 0	0 0 0
	3	2 0.400 0.400	1.225 0.350 0.087	0.816 0.272 0	0.500 0.167 0	0 0 0
	4	2 0.400 0.400	1.225 0.306 0	0.816 0.204 0	0.500 0.125 0	0 0 0
	5	2 0.400 N/A	1.225 0.245 N/A	0.816 0.163 N/A	0.500 0.100 N/A	0 0 N/A

E is the total output ($E = mx_{l_i} + (n - m)x_{d_j}$). x_{l_i} is loyal subordinate i's partial equilibrium effort level. x_{d_j} is disloyal subordinate j's partial equilibrium effort level. k is competition intensity. m is the number of loyal subordinates.

Charts and Figures

Table 2 THE SUPERIOR'S UTILITY

U_S		k = 1	2	3	4	5
m	0	2	$\frac{\sqrt{6}}{2}$	$\frac{\sqrt{6}}{3}$	$\frac{1}{2}$	0
	1	$2 - \frac{3}{5}\alpha$	$\frac{\sqrt{6}}{2}$	$\frac{\sqrt{6}}{3} + \frac{3\alpha}{7}$	$\frac{1}{2} + \frac{3\alpha}{4}$	α
	2	$2 - \frac{\alpha}{5}$	$\frac{\sqrt{6}}{2} + \frac{5\alpha}{8}$	$\frac{\sqrt{6}}{3} + \frac{14\alpha}{15}$	$\frac{1}{2} + \alpha$	α
	3	$2 + \frac{\alpha}{5}$	$\frac{\sqrt{6}}{2} + \frac{13\alpha}{14}$	$\frac{\sqrt{6}}{3} + \alpha$	$\frac{1}{2} + \alpha$	α
	4	$2 + \frac{3\alpha}{5}$	$\frac{\sqrt{6}}{2} + \alpha$	$\frac{\sqrt{6}}{3} + \alpha$	$\frac{1}{2} + \alpha$	α
	5	2α	$\frac{\sqrt{6}}{2}\alpha$	$\frac{\sqrt{6}}{3}\alpha$	$\frac{1}{2}\alpha$	α

*

References

Acemoglu, D., S. Johnson, and J. A. Robinson (2001). The colonial origins of comparative development: An empirical investigation. *American Economic Review 91*(5), 1369–1401.

Acemoglu, D., S. Naidu, P. Restrepo, and J. A. Robinson (2019). Democracy does cause growth. *Journal of Political Economy 127*(1), 47–100.

Acemoglu, D. and J. A. Robinson (2005). *Economic Origins of Dictatorship and Democracy*. Cambridge: Cambridge University Press.

Alesina, A. and E. L. Glaeser (2004). *Fighting Poverty in the US and Europe: A World of Difference*. Oxford University Press.

Andersen, D. D. E. (2018). Does meritocracy lead to bureaucratic quality? revisiting the experience of prussia and imperial and weimar germany. *Social Science History 42*(2), 245–268.

Baker, G., R. Gibbons, and K. J. Murphy (1994). Subjective performance measures in optimal incentive contracts. *The Quarterly Journal of Economics 109*(4), 1125–1156.

Bennedsen, M., K. M. Nielsen, F. Pérez-González, and D. Wolfenzon (2006). Inside the family firm: The role of families in succession decisions and performance. *ECGI-finance working paper* (132).

Berry, S. K. (1993). Rent-seeking with multiple winners. *Public Choice 77*(2), 437–443.

Bloom, N. and J. Van Reenen (2007). Measuring and explaining management practices across firms and countries. *The quarterly journal of Economics 122*(4), 1351–1408.

Bloom, N. and J. Van Reenen (2010). Why do management practices differ across firms and countries? *Journal of economic perspectives 24*(1), 203–224.

Chen, J. (2024). State formation and bureaucratization: Evidence from pre-imperial china. *The Journal of Economic History 84*(3), 690–726.

Coase, R. (1937). The nature of the firm. *Economica 4*, 386–405.

Dessein, W. and L. Garicano (2025). La cordata: Loyalty in political tournaments. Technical report, Working Paper.

Dincecco, M. (2011). *Political Transformations and Public Finances: Europe, 1650–1913*. Political Economy of Institutions and Decisions. Cambridge University Press.

Egorov, G. and K. Sonin (2011). Dictators and their viziers: Endogenizing the loyalty–competence trade-off. *Journal of the European Economic Association 9*(5), 903–930.

Gerring, J., P. Bond, W. T. Barndt, and C. Moreno (2005, April). Democracy and economic growth: A historical perspective. *World Politics 57*(3), 323–364.

Greif, A. (2006). *Institutions and the Path to the Modern Economy: Lessons from Medieval Trade*. Cambridge: Cambridge University Press.

Hobbes, T. (1968). *Leviathan*. Harmondsworth,: Penguin Books.

Holmström, B. (1982). Moral hazard in teams. *The Bell Journal of Economics 13*(2), 324–340.

Huntington, S. P. (1973). *Political Order in Changing Societies*. Yale University Press.

Jia, R., M. Kudamatsu, and D. Seim (2015). Political selection in china: The complementary roles of connections and performance. *Journal of the European Economic Association 13*(4), 631–668.

Johnson, C. (1982). *MITI and the Japanese miracle: the growth of industrial policy, 1925-1975*. Stanford university press.

Karaman, K. K. and Ş. Pamuk (2010). Ottoman state finances in european perspective, 1500–1914. *The Journal of Economic History 70*(3), 593–629.

Lazear, E. P. and S. Rosen (1981, October). Rank-order tournaments as optimum labor contracts. *Journal of Political Economy 89*(5), 841–864.

Lipset, S. M. (1960). *Political Man: the Social Bases of Politics*. Doubleday and Company, Garden City.

Lipset, S. M. and S. Rokkan (1967). Cleavage structures, party systems, and voter alignments: An introduction. In S. M. Lipset and S. Rokkan (Eds.), *Party Systems and Voter Alignments: Cross-National Perspectives*, pp. 1–64. New York: Free Press.

Mobarak, A. M. (2005). Democracy, volatility, and economic development. *Review of Economics and Statistics 87*(2), 348–361.

North, D. C. (1990). *Institutions, Institutional Change and Economic Performance*. Cambridge: Cambridge University Press.

North, D. C. (1992). Institutions, ideology, and economic performance. *Cato Journal 11*(3), 477–496.

Padgett, J. F. and P. D. McLean (2006). Organizational invention and elite transformation: The birth of partnership systems in renaissance florence. *American Journal of Sociology 111*(5), 1463–1568.

Pamuk, Ş. (2020). The ottoman empire: Institutions and economic change, 1500–1914. In A. Banerjee (Ed.), *Oxford Research Encyclopedia of Economics and Finance*. New York, NY: Oxford University Press.

Pérez-González, F. (2006). Inherited control and firm performance. *American Economic Review 96*(5), 1559–1588.

Persson, T. and G. Tabellini (2006). Democracy and development: The devil in the details. *American Economic Review 96*(2), 319–324.

Polanyi, K. (1944). *The Great Transformation: The Political and Economic Origins of Our Time*. New York: Farrar & Rinehart.

Prendergast, C. (1999). The provision of incentives in firms. *Journal of Economic Literature 37*(1), 7–63.

Prendergast, C. and R. H. Topel (1996). Favoritism in organizations. *Journal of Political Economy 104*(5), 958–978.

Przeworski, A. (2000). *Democracy and Development: Political Institutions and Well-Being in the World, 1950-1990*. Cambridge University Press.

Rai, B. K. and R. Sarin (2009, July). Generalized contest success functions. *Economic Theory 40*(1), 139–149.

Romer, P. M. (1990). Endogenous technological change. *Journal of Political Economy 98*(5), S71–S102.

Sachs, J. D. and A. M. Warner (1995). Natural resource abundance and economic growth. *NBER Working Paper* (5398).

Skaperdas, S. (1996, June). Contest success functions. *Economic Theory 7*(2), 283–290.

Solow, R. M. (1957). Technical change and the aggregate production function. *Review of Economics and Statistics 39*(3), 312–320.

Tavares, J. and R. Wacziarg (2001). How democracy affects growth. *European Economic Review 45*(8), 1341–1378.

Wang, Y. (2022). *The Rise and Fall of Imperial China: The Social Origins of State Development*. Princeton University Press.

Zakharov, A. V. (2016). The loyalty-competence trade-off in dictatorships and outside options for subordinates. *Journal of Politics 78*(2), 457–466.

Zingales, L. (2017). Towards a political theory of the firm. *Journal of Economic Perspectives 31*(3), 113–130.